

options for those who seek them. It has an easy-to-use interface and can prepare files of up to 20 microns for printing. The software is available for Windows, Ubuntu and Mac OS. It can print from SD cards and is Wi-Fi ready. Ultimaker later plans to release an upgraded model, which will have dual printing heads.

6. Michigan Technological University's open source 3D metal printer

The Michigan Technological University (MTU) has upped the ante by developing an open source 3D metal printer costing less than \$1,500 to build. This price tag is dramatically lower than commercial 3D metal printers, which can cost around half a million dollars.

Anyone can create their own 3D metal printer as MTU has made the detailed plans and software available as open source.

The 3D printer consists of a small commercial gas-metal arc welder and an open source microcontroller. Here, the extruder is made up of the welder, which melts metal wire and deposits drops of liquid metal in layers. This allows it to create complex geometric structures. In the future when this device is improved upon, it could be used to print items that range from household objects to custom lab equipment. The lab's report states that with some additions it may be possible to use aluminium as the printing material.



7. Fab@Home

'Fab@Home' is a platform of printers that can build dynamic 3D objects. Its purpose is to make real, usable objects. Case in point being a working flash-light printed by Fab@Home. The Fab@Home printer is not self-replicating like some other printers and is geared towards printing static and dynamic objects. It is also easier to put together than many other printers. This printer uses a wide variety of materials to print the object layer by layer. The material can be in liquid or paste form, as long as it can be squeezed through its syringes. Some of the materials that have been used until now include chocolate, cheese, epoxy, gypsum plaster, PlayDoh and ceramic clay mixed with water. As part of a demonstration related to printing body parts, the